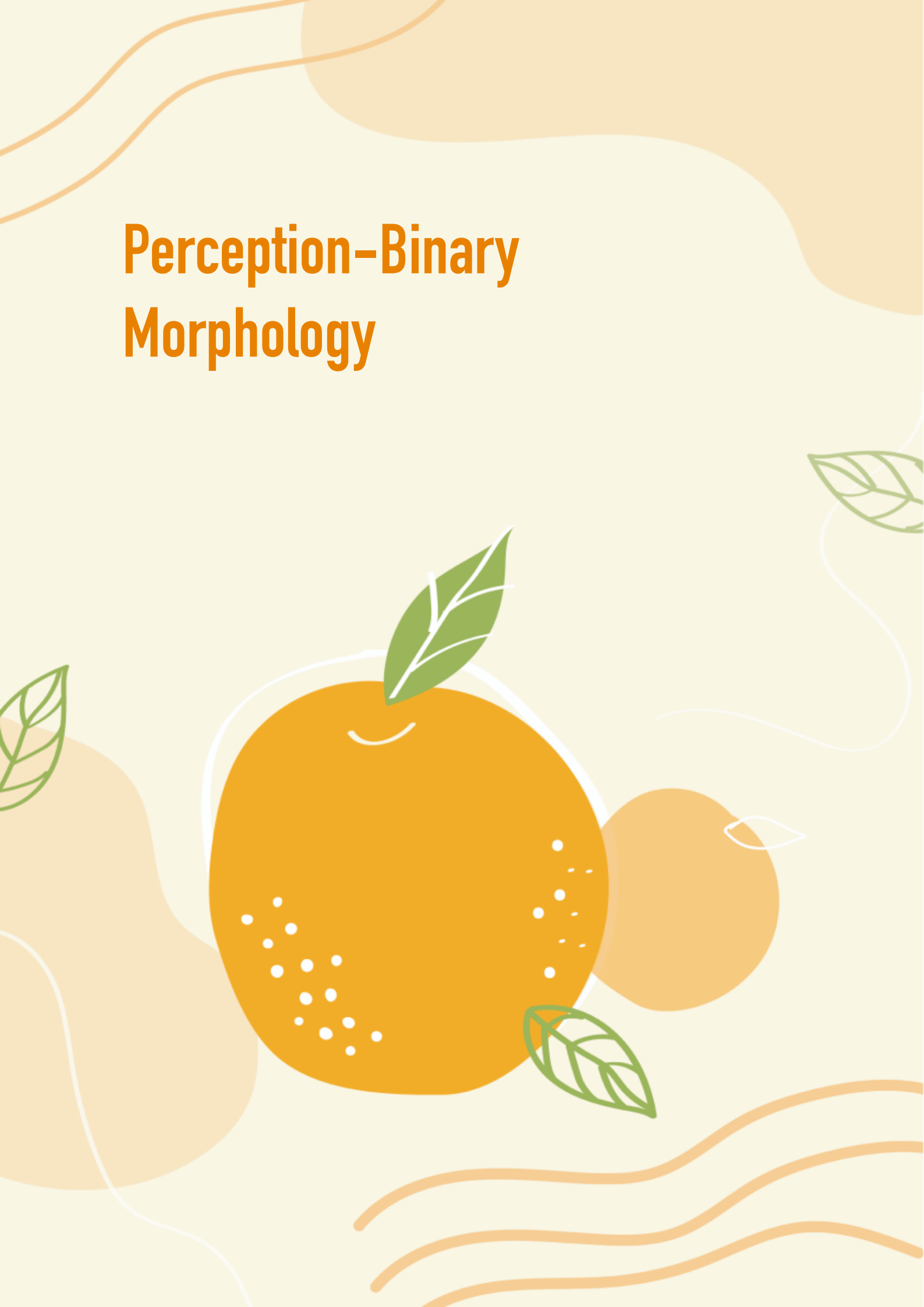


# Perception-Binary Morphology



**Binary Morphology**: image processing technique that analyzes & alters the geometric structure of binary images by probing an image with a predefined shape, remove noise & fill gaps

What kind of Operations?

> separate objects from background & from one another

> aggregate pixels for each object

> compute features for each object

pixels similar  
to a group being  
clustered

Let say you have a binary image & you need to classify objects present in that image:

Operations:

① Thresholding a gray scale image

② Determining good thresholds

③ Connected component analysis

④ Binary mathematical morphology

⑤ Finding statistics (area ---)

we represent the  
pixels values in  
the form a  
a histogram

**Thresholding**: 0-255 → set a number b/w it & all the pixels of values less than this no. = black & all the pixels of values greater than this no. = white

How to find a threshold?

Automatic thresholding: Otsu's method

assumption: bimodal  
histogram

① calculates the probability of each intensity.

② iterates through all the intensities & divide them

into 2 classes:   
 ↗ background   
 ↘ foreground

mid: set of pixels   
 ↓   
 will be the potential threshold.

③ for each of the potential threshold   
 calculate its intra & inter class variance

↙ how much pixels of each background & foreground are diff   
 ↘ how diff the background & foreground pixels are from each other

④ the potential threshold which has least intra class variance & max inter class variance will be the chosen threshold

## Connected Components Analysis:

↳ when pixels of 2 diff object are so similar then the 2 object looks connected:

Connected components method: Row-by-Row algorithm

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |

↳ this looks like one single object

steps: let a counter = 0

start from row-1, when the first '1' is encountered → counter++, then as we traverse ahead,

pass set of 0s then again '1' is encountered → counter++

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 2 | 2 | 2 | 2 | 0 | 3 | 3 | 3 | 3 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 2 | 2 | 2 | 2 | 0 | 0 | 3 | 3 | 3 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 2 | 2 | 2 | 2 | 0 | 0 | 3 | 3 | 3 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 2 | 2 | 2 | 2 | 0 | 0 | 3 | 3 | 3 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 3 | 3 | 3 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 3 | 3 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |

should be connected by 1 from all the 4 sides

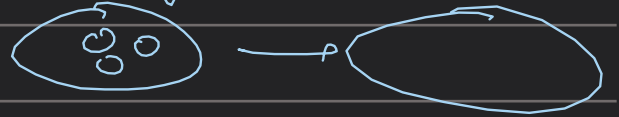
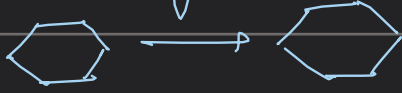
↖ conflicting points

# Mathematical Morphology.

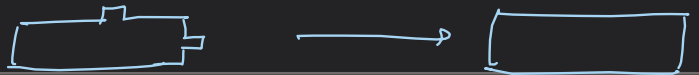
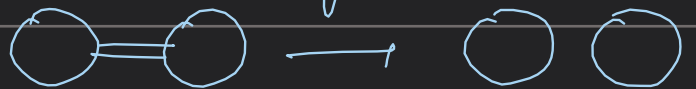
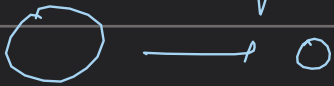
## ① Dilation

## ② Erosion

Dilation: expands the connected set of 1s of a binary image  
Used for: Growing features      Filling holes & gaps



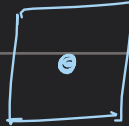
Erosion: shrinks the connected sets of 1s of a binary image  
used for: Shrinking features      Removing branches, bridges



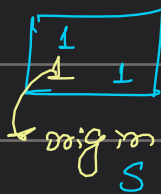
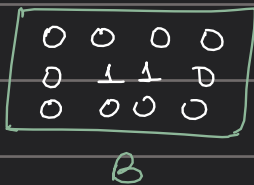
## Structuring element:

A shape mask used in basic morphological operations

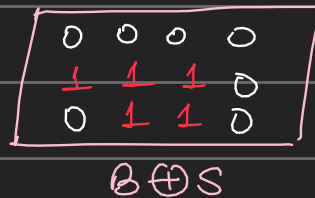
• with a defined origin



Dilation: input: Binary image  $B$ , structuring element  $S$   
• Move  $S$  over  $B$ , placing the origin at each pixel  
• Considering only 1-pixel locations in  $S$ , compute the binary **OR** of corr. elements in  $B$ .



dilate



Erosion: input: Binary image  $B$ , structuring element  $S$

• Move  $S$  over  $B$ , placing the origin at each pixel

• Considering only 1-pixel locations in  $S$ , compute the binary **AND** of corr. elements in  $B$ .

$$\begin{array}{ccccc}
 0 & 0 & 1 & 1 & 0 \\
 0 & 0 & 1 & 1 & 0 \\
 0 & 0 & 1 & 1 & 0 \\
 1 & 1 & 1 & 1 & 1
 \end{array}$$

B

$$\begin{array}{c}
 1 \\
 1 \\
 1
 \end{array}$$

S

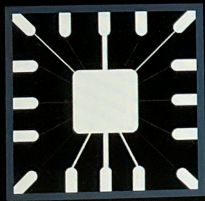
origin

erode

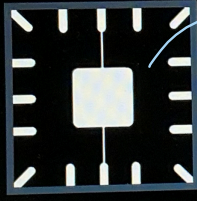
$$\begin{array}{ccccc}
 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 1 & 1 & 0 \\
 0 & 0 & 1 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0
 \end{array}$$

$B \ominus S$

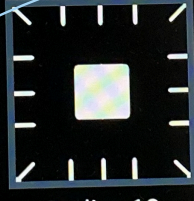
### Effect of disk size on erosion



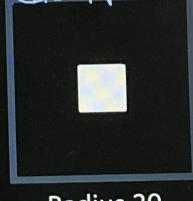
Original image



Erosion with a disk of radius 5



Radius 10



Radius 20

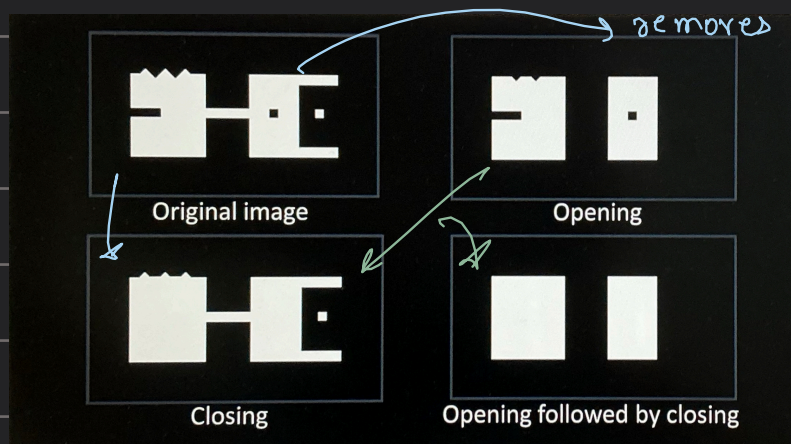
$1 \oplus 0 \rightarrow 0$   
write disappearing

Opening & closing: (erosion + dilation) for preventing change in size of the object in the binary image

Opening: a compd. operation of erosion followed by dilation (with the same structuring element)  
connected objects — creates a gap b/w them

closing: a compd. operation of dilation followed by erosion (with the same structuring element)  
disconnected object — fuses them like a bridge

fills the gap



removes the gap / bridge



## Some Basic Morphological Algorithms

### ① Boundary extraction:

The boundary of  $A$  can be computed as

$$A - (A \ominus B)$$

$B$ : 3x3 structuring element

$$\textcircled{2} \text{ Thinning: } A \otimes B = A - (A \ominus B) = A \cap (A \ominus B)^c$$

$$\textcircled{3} \text{ Thickening: } A \oplus B = A \cup (A \otimes B)$$